When Operator Accuracy Does Not Transfer to the Image: A Case Study of a Learned Interior Transport Operator for Faceted Gemstones

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***Abstract*—A diamond's appearance is made inside it: light refracts in, reflects between facets and leaves in sharply defined directions. Neural appearance models compress such internal light transport into a small network and work well for cloth, where the structure is far smaller than a pixel. This study asks whether the idea transfers to faceted gemstones, where the structure *is* the appearance. A radiance distribution map (RDM) adapted from the cloth method is first shown to condition on the wrong variable, and is generalised to a *learned interior operator*: a network that takes where, in which direction and at what wavelength light enters the stone, and predicts whether, through which facet and in which direction it leaves. Three decoders were trained to convergence on three diamond cuts with five seeds each (45 runs) and compared with an exact spectral path tracer. The deterministic decoders predict exit facets more accurately than the stochastic one on every cut, yet their images agree with the reference *less* (pixel correlation 0.050–0.184 against 0.122–0.272), in 29 of 30 paired seeds. Held-out paths locate the failure: exit facets are predicted correctly for 45–59% of paths meeting two or three facets but 2–3% of those meeting sixteen or more. Each learned decision is three times faster than tracing, yet renders are 31% slower and the neural image settles at an error a single analytic render already beats.**

***Index Terms*—neural rendering, light transport, gemstones, mixture density networks, image evaluation, Monte Carlo rendering**

I. Introduction

The appearance of a faceted diamond depends almost entirely on where light leaves it. Light that enters the crown is refracted, reflected between facets several times and refracted out again, and each facet acts as a mirror onto a different part of the surroundings. Small changes in where light enters move these reflections across facet edges, which is what produces brilliance, fire and sparkle. Path tracing reproduces this exactly but repeats the whole interior journey for every sample.

Neural appearance models offer a way to learn such journeys once and reuse them. Soh and Montazeri [1] compress the light scattered inside cloth fibres into small networks trained from a radiance distribution map (RDM): a table, gathered by tracing the true transport, of how light leaves for each incoming direction. For cloth this is well founded: hundreds of fibres sit inside one pixel, so light enters and leaves within a sub-pixel cross-section and direction alone describes it. A diamond inverts this. It has a few dozen facets, each covering thousands of pixels, and *where* light enters decides which facet it leaves through.

This study asks: *can a compact learned operator reproduce the interior light transport of a faceted gemstone, and if not, which of its errors explain the difference in the image?* Three hypotheses make the question testable:

- **H1 (dissociation).** An operator can improve its exit-facet accuracy without improving the image's agreement with the exact reference.
- **H2 (localisation).** Its errors concentrate in long internal routes and pavilion exits, not in facets that lack training data.
- **H3 (cost).** A faster interior decision does not make the image cheaper at equal quality.

All three are tested under one studio lighting rig on three procedural cuts; the conclusions concern this setting rather than all learned representations of specular media.

II. Background

A. Light transport in a diamond

When light meets a diamond's surface, part of it reflects and the rest refracts into the stone. The split is given by the Fresnel equations and the bend by Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$, with refractive index $n \approx 2.42$ for diamond. Inside, light meeting a facet beyond the critical angle of about $24.4°$ is totally internally reflected, so it may bounce many times before escaping. The index also varies with wavelength (dispersion), which splits white light into the colours known as fire. A cut is designed so that light entering the crown is reflected by the pavilion back out through the crown; the result is a pattern of sharp reflections whose positions depend sensitively on the viewpoint and on where light enters.

B. Path tracing and Monte Carlo noise

Physically based renderers estimate the light reaching each pixel by tracing random light paths [2], [3]. Each pixel averages many samples per pixel (spp); the estimate is unbiased but noisy, with variance that falls in proportion to $1/N$ for N samples. Two renders with different random seeds therefore differ even when the method is exact, and any comparison between images must account for this noise. For a gemstone, every camera path that enters the stone must be traced through

[0]Source code available at: [GitHub repository].

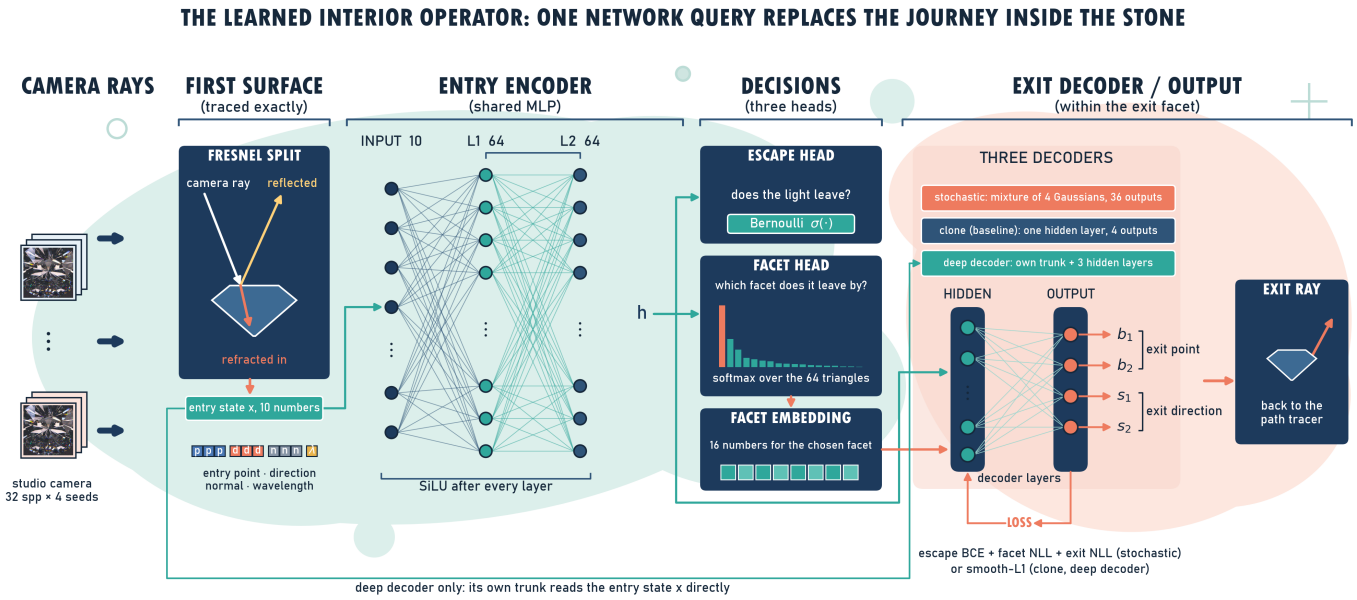


Fig. 1. The learned interior operator. Camera rays meet the stone's first surface, where Fresnel reflection or refraction is traced exactly. A refracted ray's entry state (position, direction, normal, wavelength; 10 numbers) passes through a shared two-layer encoder to an escape head, a facet head over the mesh's triangles, and a 16-number embedding of the chosen facet. One of three decoders then places the exit on that facet, and the exit ray returns to the path tracer. The deep decoder alone also reads the entry state through its own trunk.

all of its internal bounces, which is the work a learned operator aims to replace.

C. Neural appearance models and the RDM

A neural appearance model replaces part of the light transport with a small network evaluated at render time [4], [5]. In the cloth model [1], the multiple scattering inside a yarn is gathered into an RDM by firing rays at the fibre geometry and binning the light that leaves by outgoing direction for each incoming direction; small networks are fitted to the binned table and applied as a material. The present study starts from this design and adapts it to a diamond.

III. Related Work

A. Neural appearance and learned transport

Neural materials compress measured or simulated appearance into compact networks, from neural BRDFs with importance sampling [6] to multi-resolution neural materials [5] and real-time layered models [4]. These represent light that is reflected or scattered near a surface. Closer to this study, Kallweit et al. [7] replace multiple scattering in clouds with a radiance-predicting network, and Vicini et al. [8] learn where light leaves a translucent object given where it enters, using a conditional variational autoencoder and local shape descriptors. Both address diffusive media, whose exit distributions are smooth. Learned densities also serve as importance-sampling proposals [9]. This study targets the specular case these works exclude: a smooth dielectric whose exit is a discontinuous function of the entry state.

B. Rendering gemstones

Gemstone rendering has been approached with explicit optics: Sun et al. [10] model the refraction, reflection and dispersion of diamonds, and Guy and Soler [11] render gemstones in real time by tracing a bounded number of internal reflections. Spectral rendering with hero wavelengths [12] keeps dispersion affordable. These methods are exact or deliberately truncated; none learns the interior transport.

C. Representation limits of small networks

Multilayer perceptrons are biased towards smooth functions [13]. Fourier features [14] and multiresolution hash encodings [15] are standard remedies, and mixture density networks [16] represent several possible outcomes for one input, suiting an entry state that can escape by several internal routes. Moment-based filtering such as LEAN mapping [17] preserves variation that averaging would otherwise erase. Each of these is tested or adapted below.

D. Evaluating renders

Images are commonly compared with SSIM [18] or FLIP [19]. Both are sensitive to Monte Carlo noise, which matters when a learned method is compared with a noisy reference. This study instead measures agreement against the reference's own noise level, corrects correlations for noise in the manner of Spearman's disattenuation [20], and attaches block-bootstrap intervals [21], [22].

IV. Preliminary Study: The Radiance Map

The project began as a direct adaptation of the cloth model: an RDM gathered on a diamond mesh, compressed into small

networks and applied as a material on the stone's surface. Each step below removed one explanation for why this did not produce a diamond.

a) Is it a sampling problem?: Soh and Montazeri gather 50–100 million fully traced rays per yarn material until the map converges [1]. To give the RDM the same chance, it is gathered for each cut at five budgets from 2^{26} rays (67 million) to 2^{30} (1.07 billion) and rendered under the same rig (Fig. 2). If flat shading were sampling noise it would sharpen along this ladder; if it persists while the bins converge, the limit is what the map conditions on. *[Ladder renders pending.]*

b) Averaging erases the detail: Each bin stores the mean of the paths landing in it. In 99.8% of well-populated bins the standard deviation exceeds that mean (median ratio 2.53): more rays make the mean more precise but do not make the distribution narrower. A moment-based modification after LEAN mapping [17], which switches shading points on with probability $p = 1/(1 + r^2)$ at gain $1/p$ for relative spread r, restores the *amount* of sparkle but places it arbitrarily: the result reads as speckle.

c) The aim offset fixes coverage: The first gathers aimed every ray at the stone's centre, which caps the incidence angle each facet can receive: no checkpoint before the fix filled more than 41 of its 64 incident-direction bins. Firing each ray from a random point on a disc the size of the stone's silhouette (the *aim offset*), with directions stratified uniformly in $\cos\theta$ within each bin, fills all 64. Where a ray is aimed matters as much as its direction, a lesson carried into the operator's data (Sec. VII-F).

d) The missing variable is position: Asked to predict the exit triangle, a table conditioned on entry direction, as the RDM is, scores 0.068 with 32 cells and only 0.098 with 2,048, against 0.047 for the most common triangle; conditioning on the entry triangle instead reaches 0.192. Resolution is not the limit; the conditioning variable is. The learned interior operator follows: it keeps the RDM's idea of a learned interior but conditions on entry position, direction, normal and wavelength and returns a single exit rather than a binned average.

V. Methodology

A. Scene and reference

All renders use one studio scene: a perspective camera with a 30° field of view, a grey diffuse ground plane, textured area lights and two small intense sources. Structured lighting is essential, because under uniform lighting every exit direction returns the same radiance and errors become invisible. The *analytic reference* is a spectral path tracer in Mitsuba 3 [23] that treats the stone as a smooth dispersive dielectric (Sellmeier index), samples Fresnel reflection or refraction at every interaction and traces each interior path until escape or a depth of 128. Absorption is not modelled.

B. The learned interior operator

The entry state $\mathbf{x} \in \mathbb{R}^{10}$ holds position $\mathbf{p}$ (scaled by the mesh radius), interior direction $\boldsymbol{\omega}$, entry normal $\mathbf{n}$ and wavelength λ. With e the event that light escapes before the depth cap, f the exit triangle and $\mathbf{y} \in \mathbb{R}^4$ the exit coordinates, the operator models

$$p(e = 1, f, \mathbf{y} \mid \mathbf{x}) = s_\theta(\mathbf{x})\, \pi_\theta(f \mid \mathbf{x})\, q_\theta(\mathbf{y} \mid \mathbf{x}, f), \quad (1)$$

so non-escape is explicit rather than an artificial exit. Two coordinates are barycentric logits and two are tangent slopes on triangle f:

$$\boldsymbol{\beta} = \mathrm{softmax}(y_1, y_2, 0), \quad \boldsymbol{\omega}_{\mathrm{out}} = \frac{\mathbf{n}_f + y_3\mathbf{t}_f + y_4\mathbf{b}_f}{\|\mathbf{n}_f + y_3\mathbf{t}_f + y_4\mathbf{b}_f\|}, \quad (2)$$

with exit point $\sum_j \beta_j \mathbf{v}_{fj}$, so every prediction lies on the chosen triangle and points outward.

Fig. 1 shows the architecture. A two-layer, width-64 SiLU encoder [24] feeds an escape head and a facet head; a learned 16-number embedding of the chosen facet conditions the exit decoder. Three decoders share everything else: the *stochastic* decoder is a mixture of four Gaussians [16] sampled at render time (17,637 parameters on the round cut); the *clone* regresses a single exit with a smooth-L1 loss, after behavioural cloning [25] (15,557); and the *deep clone* uses three hidden layers and its own encoder (28,741). All three sample the exit facet; they differ only in how the exit is placed on it. Training minimises

$$\mathcal{L}_i = \mathrm{BCE}(s_\theta(\mathbf{x}_i), e_i) + e_i\,[-\log \pi_\theta(f_i \mid \mathbf{x}_i) + \ell_{\mathrm{coord}}(\mathbf{y}_i)] \quad (3)$$

with Adam [26].

C. Rendering with the operator

The first surface is traced exactly in both modes; only the journey inside the stone differs. When a camera path reaches the stone, reflection or refraction is sampled from the Fresnel coefficient. A reflected path continues outside as usual. A refracted path forms its entry state and joins a batch of up to 512 waiting entries, so the network is queried efficiently. For each entry the operator samples whether light escapes, the exit facet and the exit coordinates; the path then continues from the decoded exit as an ordinary scene ray. Non-escaping paths end, and exits that would immediately re-enter the stone are discarded. In analytic mode the same entry is instead traced through every internal bounce.

VI. Experiments

A. Experimental design

Six experiments test the hypotheses.

- **E1 Decoder comparison** (H1): the stochastic, clone and deep decoders on each of the three cuts, each trained with five seeds (45 runs), all trained to convergence on the same pool per cut, tested on an independent pool and rendered against one analytic reference per cut.
- **E2 Diagnostics** (H1): *true-facet* renders, in which the exact tracer supplies each path's exit facet to the seed-23 models; and, from earlier single-seed experiments on a camera-matched pool, lookup baselines, a nearest-neighbour oracle and an input ablation.

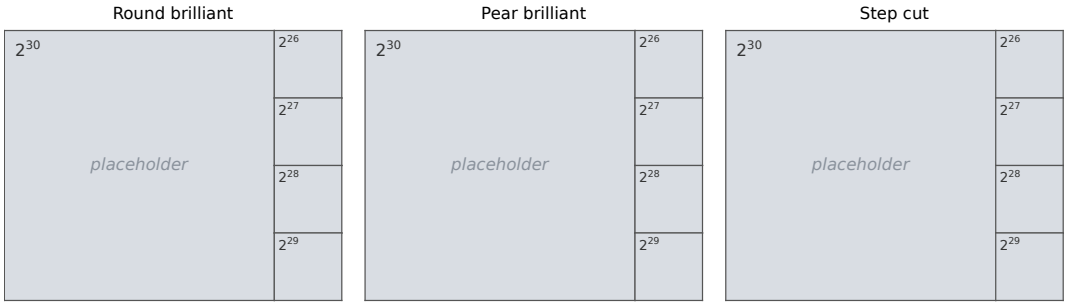


Fig. 2. RDM renders against gathering budget, after the 50–100 million rays per material of the cloth model [1]. For each cut the large panel uses 2^{30} rays and the stacked panels 2^{26} to 2^{29}; all use the same studio rig and render settings. *Placeholder layout: renders pending.*

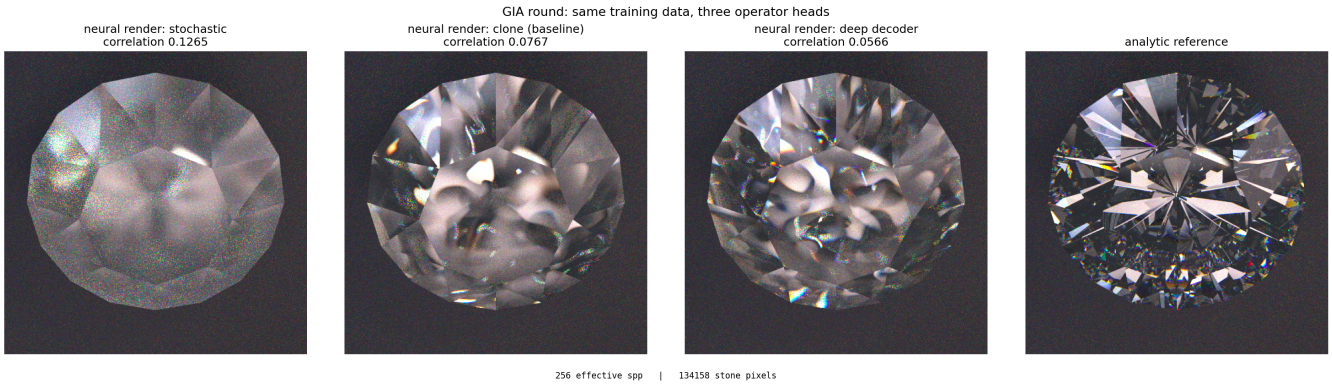


Fig. 3. Round-cut renders from E1 (seed 23): stochastic, clone and deep decoder beside the analytic reference (256 spp). Deterministic placement restores sharp features but not their locations. Panel scores are that run's pixel correlation; Table II gives five-seed means.

- **E3 Breakdown** (H2): E1's held-out paths grouped by the number of facets met inside the stone and by exit region.
- **E4 Cost** (H3): render and per-decision timing at equal time and equal quality, and the network's cost against batch size.
- **E5 Entry sampling**: camera-matched against uniform aim-offset training entries, on a 60-frame pear tumble.
- **E6 Transfer**: an operator trained on one cut, re-pointed at another without retraining: round (GIA) to pear, and pear to step cut.

B. Baselines

- **Analytic reference**: the exact path tracer at the same settings.
- **Lookup tables**: the most common exit triangle, and tables keyed on entry direction (the RDM's conditioning) or entry triangle.
- **Nearest-neighbour oracle**: the exits of the closest training entry states, a non-parametric test of whether the exit is predictable at all.

C. Evaluation metrics

Metrics use the pixels whose camera ray hits the stone, on linear luminance. Eight seeds per mode are split into two independent halves; the reference compared with its own other half gives the *noise floor*, the agreement sampling noise alone allows.

- **Pixel correlation** $\rho(N, A)$ between neural and reference halves: is light in the right place?
- **Disattenuated correlation** $\rho/\sqrt{r_N r_A}$, dividing by each image's half-split reliability r: agreement with noise removed.
- **Facet correlation** across visible facet means: is coarse brightness right?
- **Energy error** $\epsilon_E = \sum N_p / \sum A_p - 1$: is the total right?
- **Contrast**, standard deviation over mean (noise-free version from the covariance of halves): is the image as sharp as the reference?
- **Interior-only** versions of each, using only light that passed through the operator (or, for the reference, the traced interior).
- **Operator metrics** on held-out paths: exit-facet top-1 accuracy, and the exit-direction angle when the true facet is supplied.

Decoders are compared seed by seed: each deterministic run is paired with the stochastic run of the same seed, and a difference counts as clear when its 95% block-bootstrap interval excludes zero.

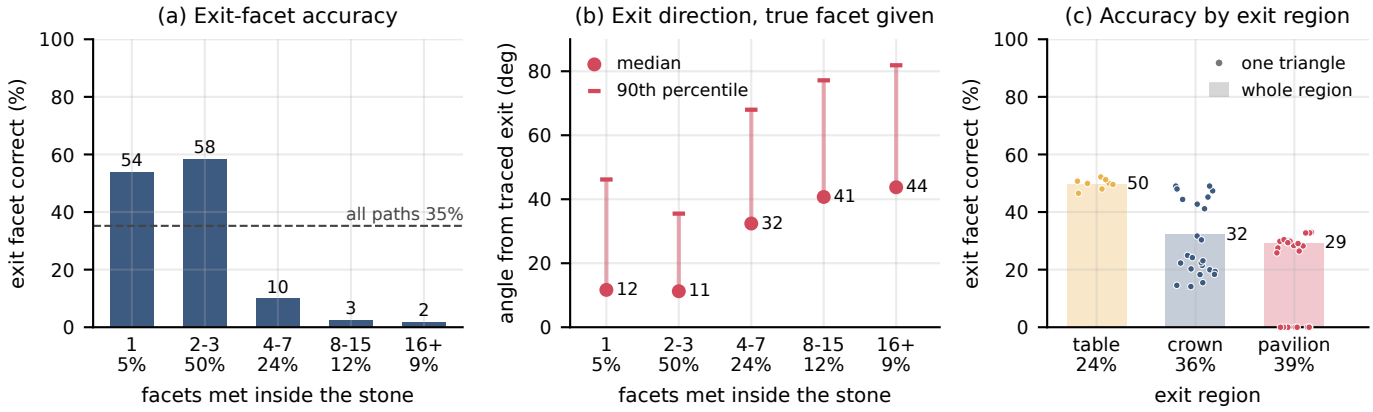


Fig. 4. E3, round cut, stochastic decoder, seed 23: 261,335 escaped held-out paths. (a) Exit-facet top-1 accuracy by the number of facets a path meets inside the stone; the second tick line is each group's share of paths. (b) Exit-direction error when the true facet is supplied: median and 90th percentile. (c) Share of exits through each region that the operator also sends there; bar: whole region, dot: one triangle with at least 20 test exits.

TABLE I
EXPERIMENTAL PARAMETERS

Parameter	Value
Cuts (triangles / facets)	round 64/57, pear 62/33, step 112/50
Training pool per cut	1,048,576 paths from 65,536 entry states: half camera-matched, half uniform with the aim offset; split 80/20 by entry state
Held-out test pool per cut	262,144 paths from 16,384 entry states, same mix, separate seeds
Interior depth cap	128
Training	Adam, learning rate 10^{-3}, batch 256, clip 10
Stopping	best validation loss improving by $\leq 10^{-3}$ for 5 epochs (stopped after 31–93)
Training seeds	23, 24, 25, 26, 27
Renders	512×512, box filter, 8×32 spp per mode (halves of 128 spp)
Operator batch	512 entry states
Hardware	14-core Linux nodes of a Slurm cluster
Software	Mitsuba 3.9.1, Dr.Jit 1.5.0, PyTorch 2.13, Python 3.10

D. Data collection

Each of the 45 runs writes, under one folder: the exact command and git commit, the software and hardware, hashes of the mesh, training data and checkpoint, per-epoch training and validation losses, per-seed render timing, every evaluation result as JSON and CSV, and its figures. A final step aggregates the five seeds of every cut and decoder and computes the paired differences. The runs were executed as Slurm array jobs; every job skips finished work, so interrupted jobs resume without repeating it.

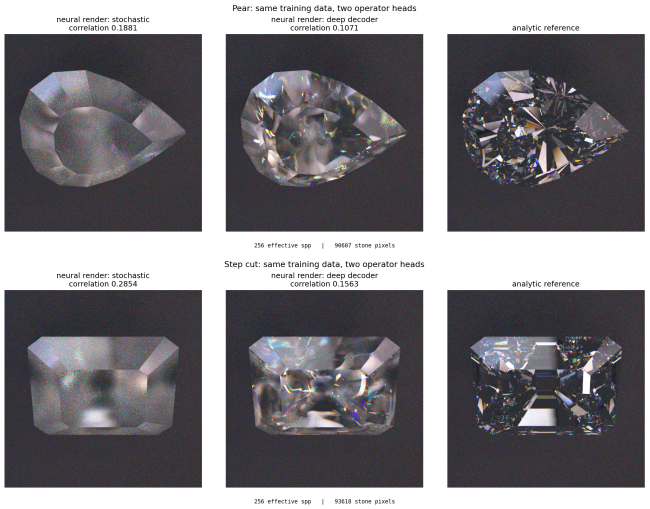


Fig. 5. Pear (top) and step cut (bottom) from E1 (seed 23): stochastic and deep-decoder operators beside their references (256 spp). Broad shape and brightness survive on both cuts; the sharp internal reflections do not.

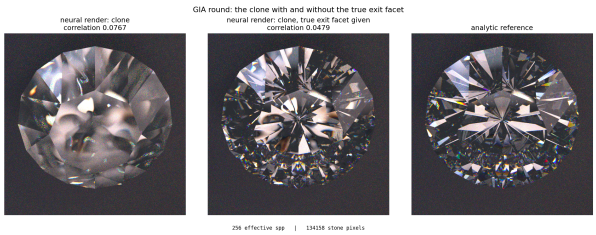


Fig. 6. E2, round cut, seed 23: the clone, the clone supplied with the traced exit facet, and the reference (256 spp). The correct facet restores the star pattern and contrast but not placement.

VII. Results and Analysis

A. The operator learns (E1, E2)

a) Learning beats every lookup: All three decoders predict the round cut's exit triangle better than every lookup baseline and the nearest-neighbour oracle (Table III), with a

TABLE II

IMAGE AGREEMENT IN E1: MEAN ± STANDARD DEVIATION OVER FIVE TRAINING SEEDS. EACH RENDER IS 8×32 SPP; PIXEL, DISATTENUATED AND INTERIOR-ONLY CORRELATIONS COMPARE 128 SPP HALVES, FACET CORRELATION AND NOISE-FREE CONTRAST USE THE FULL AVERAGE. FLOORS: THE ANALYTIC REFERENCE AGAINST ITS OWN OTHER HALF.

Cut	Decoder	Pixel corr. ↑	Disattenuated ↑	Interior only ↑	Facet corr. ↑	Energy error	Contrast
Round	Stochastic	**0.122** ± 0.015	**0.154** ± 0.024	**0.057** ± 0.014	**0.260** ± 0.064	+8% ± 2	0.85 ± 0.09
	Clone	0.050 ± 0.025	0.053 ± 0.027	0.021 ± 0.017	0.250 ± 0.122	+21% ± 18	2.22 ± 0.54
	Deep decoder	0.060 ± 0.037	0.063 ± 0.038	0.032 ± 0.037	0.120 ± 0.076	+13% ± 12	2.69 ± 0.71
	floor / reference	*0.974*			*0.9999*		*3.72*
Pear	Stochastic	**0.172** ± 0.014	**0.281** ± 0.020	**0.103** ± 0.016	**0.690** ± 0.072	+19% ± 2	0.48 ± 0.02
	Clone	0.074 ± 0.038	0.081 ± 0.042	0.034 ± 0.042	0.476 ± 0.214	+46% ± 17	1.74 ± 0.26
	Deep decoder	0.092 ± 0.019	0.103 ± 0.022	0.050 ± 0.018	0.451 ± 0.138	+40% ± 5	1.46 ± 0.22
	floor / reference	*0.941*			*0.9999*		*1.44*
Step	Stochastic	**0.272** ± 0.018	**0.356** ± 0.021	**0.197** ± 0.020	**0.857** ± 0.023	+16% ± 3	0.59 ± 0.02
	Clone	0.112 ± 0.036	0.121 ± 0.040	0.058 ± 0.027	0.713 ± 0.087	+29% ± 9	1.53 ± 0.36
	Deep decoder	0.184 ± 0.026	0.203 ± 0.031	0.124 ± 0.026	0.852 ± 0.031	+15% ± 4	1.09 ± 0.14
	floor / reference	*0.975*			*0.9999*		*2.35*

TABLE III

ROUND-CUT EXIT-TRIANGLE PREDICTION ON HELD-OUT PATHS. UPPER ROWS: EARLIER SINGLE-SEED DIAGNOSTICS ON A CAMERA-MATCHED POOL; LOWER ROWS: E1, MEAN ± STANDARD DEVIATION OVER FIVE SEEDS ON THE MIXED TEST POOL.

Predictor	Top-1 accuracy
Most common triangle	0.047
Entry-direction table (32 / 2,048 cells)	0.068 / 0.098
Entry-triangle table	0.192
Nearest neighbour / majority of eight	0.287 / 0.343
Stochastic decoder	0.358 ± 0.005
Clone	0.386 ± 0.005
Deep decoder	0.386 ± 0.003

TABLE IV

E2: SUPPLYING THE TRUE EXIT FACET TO THE SEED-23 MODELS. PIXEL CORRELATION AND NOISE-FREE CONTRAST, WITHOUT AND WITH THE TRUE FACET. STEP CUT PENDING.

		Pixel corr.		Contrast (ref.)	
Cut	Decoder	learned	true facet	learned	true facet
Round	Stochastic	0.127	0.151	0.79	0.98 (3.72)
	Clone	0.077	0.048	1.73	2.73 (3.72)
	Deep	0.057	0.077	1.53	2.50 (3.72)
Pear	Stochastic	0.188	0.310	0.50	0.84 (1.44)
	Clone	0.121	0.127	1.94	4.90 (1.44)
	Deep	0.107	0.107	1.41	3.38 (1.44)

spread of at most 0.005 between seeds. Training converged after 31–93 epochs, so the networks are not under-trained.

b) Position carries the information: In the earlier ablation, removing position from the input dropped top-1 accuracy from 0.404 to 0.273, and an operator given only entry direction, the RDM's conditioning, fell to 0.117. Removing wavelength changed it by 0.0003.

TABLE V

E4 COST, ROUND CUT, STOCHASTIC DECODER, SEED 23: 512×512, 32 SPP PER RENDER, MEAN OF EIGHT SEEDS. ERROR IS RELATIVE RMSE AGAINST THE CONVERGED ANALYTIC IMAGE.

	Analytic	Neural
Render time (s)	**264**	345
Time per interior decision (μs)	31	**10.5**
Interior share of render time	50%	13%
Error at equal time (2,760 s)	**0.36**	3.72
Lowest reachable error	$\to 0$	3.69
Time to reach the other's best	**26 s**	never
Training time	–	485 s
Model size	1 KB mesh	70 KB

B. Sharper is not more faithful (E1, H1)

a) A more accurate operator agrees less: On every cut the clone and deep decoders predict exit facets more accurately than the stochastic decoder on the held-out pool (round 0.386 against 0.358, pear 0.31 against 0.27, step 0.29 against 0.24), and their images look more like a diamond (Figs. 3 and 5). Yet their pixel correlation is lower on every cut (Table II). Paired by seed, the deterministic decoder is below the stochastic one in 29 of 30 pairs, and clearly so in 26. This supports H1.

b) The gap is not noise: Removing sampling noise (disattenuated correlation) leaves the ordering unchanged, and every score stays far below the noise floors of 0.94–0.98. Light that actually passes through the network agrees least: interior-only correlations are about half the whole-stone values.

c) The stochastic image is right on average, wrong in detail: The stochastic decoder keeps the best agreement but a quarter to a third of the reference's contrast (0.85 against 3.72 on the round cut): its exits spread so widely that each pixel averages bright and dark parts of the lighting.

d) Coarse structure can survive: Averaged over facets, the step cut's deep decoder matches the stochastic one (0.852 against 0.857) while its pixel correlation is lower: which facets are bright can be right while the sparkle within them is wrong.

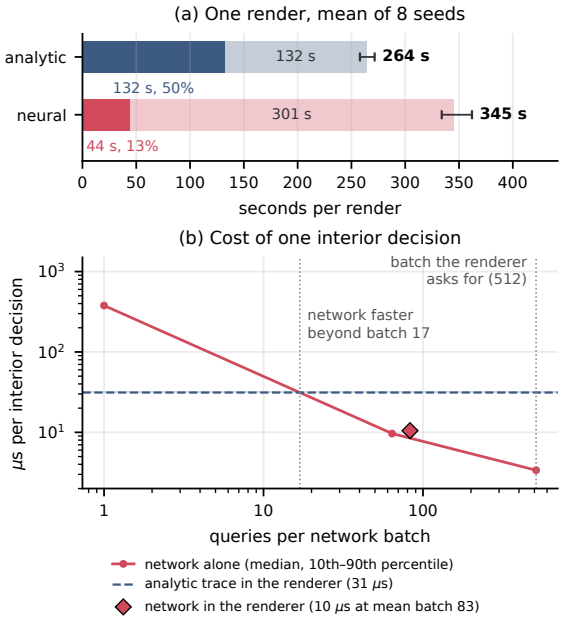


Fig. 7. E4, where the time goes. (a) Render time split into interior decisions and everything else, with the seed range. (b) Cost of one interior decision against batch size: the network alone (curve), the analytic trace in the renderer (dashed) and the network in the renderer at its mean batch (diamond).

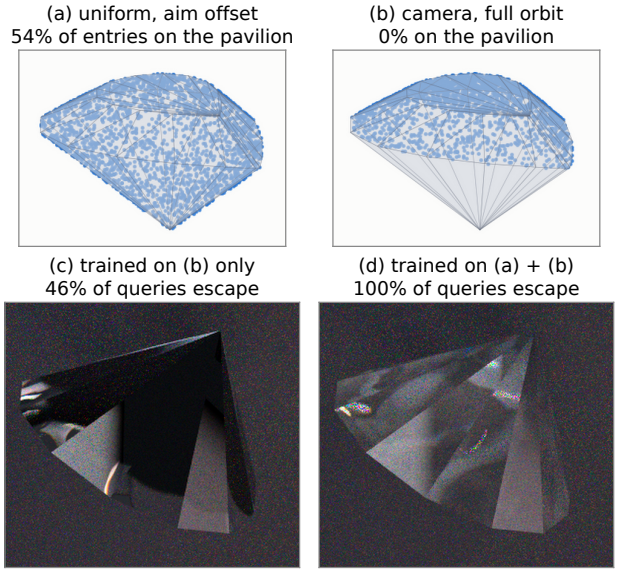


Fig. 8. E5, entry sampling on the pear. Top: where training entries land under (a) uniform directions with the aim offset and (b) camera rays from a full orbit. Bottom: a pavilion-facing tumble pose (frame 53 of 60, 256 spp) rendered by operators trained on (b) alone and on (a) and (b) together.

C. Choosing the facet is not enough (E2)

a) The right facet restores the look: Supplying the true exit facet raises the deterministic decoders' contrast towards or beyond the reference's (Table IV, Fig. 6): on the pear the clone reaches 4.90 against 1.44.

b) The right facet does not fix agreement: Their pixel correlation stays between 0.05 and 0.13, and the round clone's even falls (0.077 to 0.048), because the exit is still misplaced within the facet. Only the stochastic decoder gains clearly (pear 0.188 to 0.310), as its broad placement keeps overlap once the facet is right. Facet choice and within-facet placement are therefore separate failures.

D. Where the operator fails (E3, H2)

a) Accuracy collapses on long paths: On all three cuts, paths meeting two or three facets are predicted correctly 45–59% of the time and leave within a median 9.5°–11° of the traced direction, but paths meeting sixteen or more are predicted correctly 2–3% of the time and miss by 44°–48° (Fig. 4a,b). This part of H2 holds on every cut and every seed.

b) The pavilion is not the problem once it is covered: H2 also predicted failure at pavilion exits. Trained on the mixed pool, pavilion exits are predicted as well as crown exits (seed 23: 29%, 27% and 28% against 32%, 32% and 16% on the round, pear and step cuts; Fig. 4c). An earlier camera-only model predicted only 6% of pavilion exits correctly, so that failure was a coverage effect (E5), not a property of the pavilion. H2 is supported for route length but not for exit region.

E. Faster per decision, slower per image (E4, H3)

a) Each decision is faster: One learned interior decision costs 10.5 µs against 31 µs to trace it, three times faster, and the interior's share of the render drops from 50% to 13% (Table V, Fig. 7a).

b) The whole render is slower: A neural render takes 31% longer: the time outside the interior grows from 132 s to 301 s, spent gathering entry states, preparing inputs and decoding exits. Batch size limits the saving (Fig. 7b): one query alone costs about 0.4 ms, the network only overtakes tracing beyond about 17 queries per batch, and although the renderer asks for 512, its batches average 83 as paths finish.

c) No budget reaches analytic quality: The neural error is a fixed bias plus noise and levels off at 3.69, whereas the analytic image reaches that error after about 26 s. At equal time the analytic error is 10 times lower on the round cut and 7–10 times lower across the cuts. This supports H3.

F. Entry sampling decides coverage (E5)

a) Camera sampling misses the pavilion: Camera-matched entries match what the renderer asks, but a camera above the stone sees only the crown, so no entries land on the pavilion; uniform sampling with the RDM's aim offset puts 54% there (Fig. 8). Trained on camera entries alone, the operator lets only 46% of queries escape when the pavilion faces the camera, and the pavilion renders nearly black.

b) Coverage repairs escape, not placement: Adding uniform aim-offset entries restores escape to 100% and lights the pavilion. In earlier tests the correlation at that pose stayed at 0.018 against a floor of 0.951. This is why every E1 pool mixes the two strategies.

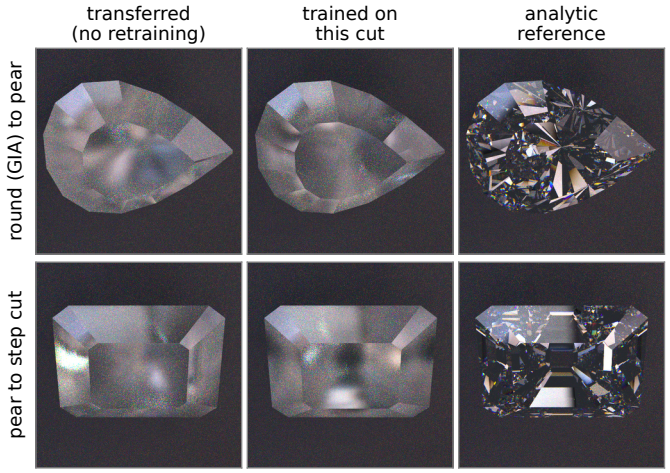


Fig. 9. E6: an operator trained on one cut and re-pointed at another without retraining, beside the stochastic operator trained on the target cut and the analytic reference (256 spp).

G. Transfer across cuts (E6)

a) What transfer requires: The operator's facet head is a categorical over the triangles of its training mesh, so each target triangle is given the identity of the source triangle whose outward normal points most nearly the same way, and the head's weights and facet embeddings are permuted to match; nothing is retrained. Round to pear (64 to 62 triangles) drops two source classes; pear to step cut (62 to 112) lets 50 surplus triangles share the probability of their best-aligned source triangle. Matching by normal ignores position, and the network keeps the internal routes of the source stone.

b) Round to pear partly transfers: The seed-23 round operator on the pear (normals matched within a median $7.8°$, mean $30.9°$) keeps much of the coarse facet brightness (facet correlation 0.553 against 0.690 for the pear's own operator), but its pixel correlation falls from 0.172 to 0.070, its interior-only correlation to -0.005, and it adds 44% too much energy.

c) Pear to step cut does not: Across families the transfer fails outright: pixel correlation -0.018, interior-only -0.040 and facet correlation 0.001, against 0.272, 0.197 and 0.857 for the step cut's own operator. Learned routes do not carry over to a different cut family. *[Deep-decoder transfers pending.]*

VIII. Discussion

A. Why accuracy does not reach the image

For entry state $\mathbf{x}$ and exit state z, the light carried back is

$$L(\mathbf{x}) = \int L_{\text{ext}}(z)\, p(z \mid \mathbf{x})\, \mathrm{d}z. \tag{4}$$

The training loss weights every exit equally, but the image weights each exit by the radiance L_{ext} it reaches: a small angular error at the edge of a bright lamp matters greatly, a large one between dark directions hardly at all. Facet accuracy and mean angular error are therefore incomplete proxies for the image. A broad exit distribution spreads energy over correct and incorrect regions and keeps some overlap; a sharp but misplaced one loses it, which is why the sharper decoders score lower in E1 and gain least from the true facet in E2. This explanation is consistent with the results but untested; a radiance-weighted objective would test it.

E3 adds that accuracy follows route length on every cut once coverage is fixed. A long route changes facet whenever the path crosses any internal facet edge, so its exit varies rapidly with the entry state, and a small smooth network blends across such changes [13]. Local encodings [15] are a natural next test.

B. Limitations

The study uses one lighting rig, one view for the main comparison, simplified meshes and a facet head tied to its mesh. Five training seeds bound the seed-to-seed spread but leave the intervals wide. The lookup baselines, nearest-neighbour oracle and ablation come from earlier single-seed runs on a camera-matched pool. The step-cut true-facet renders, the deep-decoder transfers and the RDM ladder are pending, and ground caustics are not evaluated.

IX. Conclusion

This study adapted a neural appearance model from cloth to faceted gemstones and generalised its direction-conditioned radiance map into a position-aware learned interior operator, evaluated in 45 converged runs.

The operator learns: all three decoders predict exit facets better than every lookup baseline, with a seed spread of at most 0.007.

Accuracy does not transfer to the image (H1). The more accurate deterministic decoders render sharper diamonds that agree with the reference less on every cut (0.050–0.184 against 0.122–0.272), in 29 of 30 paired seeds, and the true exit facet restores their contrast but not their agreement.

The failure follows route length (H2, in part). Paths meeting two or three facets are predicted correctly about half the time, those meeting sixteen or more almost never; pavilion exits fail only when the pavilion is missing from training.

The operator does not save time (H3). Each decision is three times faster, yet renders are 31% slower and the image settles at an error the analytic renderer reaches in under half a minute.

Future work should test radiance-weighted training and local encodings on the long-route cases, and repeat the study under other lighting.

References

[1] G. Y. Soh and Z. Montazeri, "Neural appearance model for cloth rendering," *Computer Graphics Forum*, vol. 43, no. 4, 2024, doi: 10.1111/cgf.15156.

[2] J. T. Kajiya, "The rendering equation," *Computer Graphics (Proc. SIGGRAPH)*, vol. 20, no. 4, pp. 143–150, 1986.

[3] M. Pharr, W. Jakob, and G. Humphreys, *Physically Based Rendering: From Theory to Implementation*, 4th ed. MIT Press, 2023.

[4] T. Zeltner *et al.*, "Real-time neural appearance models," *ACM Trans. Graph.*, vol. 43, no. 3, 2024.

[5] A. Kuznetsov *et al.*, "NeuMIP: Multi-resolution neural materials," *ACM Trans. Graph.*, vol. 40, no. 4, 2021.

[6] A. Sztrajman, G. Rainer, T. Ritschel, and T. Weyrich, "Neural BRDF representation and importance sampling," *Computer Graphics Forum*, vol. 40, no. 6, 2021.

[7] S. Kallweit, T. Müller, B. McWilliams, M. Gross, and J. Novák, "Deep scattering: Rendering atmospheric clouds with radiance-predicting neural networks," *ACM Trans. Graph.*, vol. 36, no. 6, 2017.

[8] D. Vicini, V. Koltun, and W. Jakob, "A learned shape-adaptive subsurface scattering model," *ACM Trans. Graph.*, vol. 38, no. 4, Art. 126, 2019.

[9] T. Müller, B. McWilliams, F. Rousselle, M. Gross, and J. Novák, "Neural importance sampling," *ACM Trans. Graph.*, vol. 38, no. 5, 2019.

[10] Y. Sun, F. D. Fracchia, and M. S. Drew, "Rendering diamonds," in *Proc. 11th Western Computer Graphics Symposium*, 2000.

[11] S. Guy and C. Soler, "Graphics gems revisited: Fast and physically-based rendering of gemstones," *ACM Trans. Graph.*, vol. 23, no. 3, pp. 231–238, 2004.

[12] A. Wilkie, S. Nawaz, M. Droske, A. Weidlich, and J. Hanika, "Hero wavelength spectral sampling," *Computer Graphics Forum*, vol. 33, no. 4, pp. 123–131, 2014.

[13] N. Rahaman *et al.*, "On the spectral bias of neural networks," in *Proc. Int. Conf. Machine Learning*, 2019, pp. 5301–5310.

[14] M. Tancik *et al.*, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 7537–7547.

[15] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Trans. Graph.*, vol. 41, no. 4, Art. 102, 2022.

[16] C. M. Bishop, "Mixture density networks," Aston University, Tech. Rep. NCRG/94/004, 1994.

[17] M. Olano and D. Baker, "LEAN mapping," in *Proc. ACM SIGGRAPH Symp. Interactive 3D Graphics and Games*, 2010, pp. 181–188.

[18] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004.

[19] P. Andersson *et al.*, "FLIP: A difference evaluator for alternating images," *Proc. ACM Comput. Graph. Interact. Tech.*, vol. 3, no. 2, Art. 15, 2020.

[20] C. Spearman, "The proof and measurement of association between two things," *American Journal of Psychology*, vol. 15, no. 1, pp. 72–101, 1904.

[21] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

[22] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[23] W. Jakob *et al.*, "Mitsuba 3 renderer," 2022. [Online]. Available: https://mitsuba-renderer.org

[24] S. Elfwing, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural Networks*, vol. 107, pp. 3–11, 2018.

[25] D. A. Pomerleau, "ALVINN: An autonomous land vehicle in a neural network," in *Advances in Neural Information Processing Systems*, 1989.

[26] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learning Representations*, 2015.